Here's a **comprehensive list of commonly used AI libraries** that IT companies use **day-to-day** in **real-world AI/ML projects** — categorized by use case for clarity 

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 **Core Machine Learning Libraries**

| Library | Language | Common Use |
|---|---|---|
| **Scikit-learn** | Python | Classic ML algorithms (Regression, SVM, Clustering, PCA) |
| **XGBoost** | Python, R, Java | Gradient boosting for high-performance ML models |
| **LightGBM** | Python, C++, R | Fast gradient boosting, widely used in Kaggle/industry |
| **CatBoost** | Python, C++, R | Gradient boosting with categorical features |
| **MLlib (Apache Spark)** | Scala, Java, Python | Distributed ML for big data environments |

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 **Deep Learning Libraries**

| Library | Language | Common Use |
|---|---|---|
| **TensorFlow** | Python, C++, Java | Deep learning models for production-scale AI |
| **Keras** | Python | High-level API for TensorFlow (quick model prototyping) |
| **PyTorch** | Python, C++ | Deep learning, NLP, computer vision (research + production) |
| **Theano** | Python | Mathematical computation for DL (older, but foundational) |
| **MXNet** | Python, Scala, Julia | Used in AWS for deep learning workloads |
| **JAX (by Google)** | Python | Accelerated ML computation, auto-differentiation |

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### □ Natural Language Processing (NLP)

| Library | Language | Common Use |
|---|---|---|
| **Hugging Face Transformers** | Python | Pretrained transformer models (BERT, GPT, T5) |
| **spaCy** | Python | Industrial NLP (tokenization, NER, POS tagging) |
| **NLTK** | Python | Classic NLP and linguistic processing |
| **Gensim** | Python | Topic modeling, Word2Vec, document similarity |
| **SentenceTransformers** | Python | Sentence embeddings and semantic search |
| **Flair** | Python | NLP sequence tagging and embeddings |
| **OpenAI API / GPT Models** | API | Generative AI, summarization, Q&A |

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### □ Computer Vision

| Library | Language | Common Use |
|---|---|---|
| **OpenCV** | C++, Python, Java | Image/video processing, object detection |
| **TensorFlow Object Detection API** | Python | Pretrained object detection models |
| **YOLO (You Only Look Once)** | Python | Real-time object detection |
| **Detectron2 (Meta AI)** | Python | Instance segmentation, object detection |
| **MMDetection** | Python | OpenMMLab framework for CV |
| **Mediapipe (Google)** | Python, C++ | Face, hand, body pose tracking |
| **Albumentations** | Python | Image augmentation for training CV models |

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### □ Generative AI

| Library | Language | Common Use |
|---|---|---|
| **Diffusers (Hugging Face)** | Python | Stable Diffusion and generative image models |
| **Stable-Baselines3** | Python | Reinforcement learning for generative tasks |
| **GAN Libraries (TensorFlow, PyTorch)** | Python | Image synthesis, data generation |
| **VQGAN+CLIP** | Python | Text-to-image generation |
| **LangChain** | Python | Building AI agents, LLM workflows |
| **LlamaIndex (GPT Index)** | Python | Knowledge-based retrieval with LLMs |

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### □ Data Processing and Engineering

| Library | Language | Common Use |
|---|---|---|
| **NumPy** | Python | Numerical operations and arrays |
| **Pandas** | Python | Data manipulation, cleaning, analytics |
| **Dask** | Python | Parallel data processing |
| **Vaex** | Python | Out-of-core DataFrames for big data |
| **PyArrow** | Python | Data serialization and columnar memory format |
| **Koalas / PySpark** | Python | Big data analytics using Spark DataFrames |

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### □ Reinforcement Learning

| Library | Language | Common Use |
|---|---|---|
| **OpenAI Gym** | Python | RL environment testing and prototyping |
| **Stable Baselines3** | Python | RL algorithms (DQN, PPO, A2C) |

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|---|---|---|
| **RLlib (Ray)** | Python | Scalable distributed RL framework |
| **TensorForce** | Python | TensorFlow-based RL framework |

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### □ Multi-Modal (Text + Image + Audio)

| Library | Language | Common Use |
|---|---|---|
| **CLIP (OpenAI)** | Python | Image-text similarity and retrieval |
| **BLIP / BLIP-2** | Python | Vision-language models |
| **Whisper (OpenAI)** | Python | Speech-to-text transcription |
| **SpeechBrain** | Python | Speech recognition and processing |
| **DeepSpeech (Mozilla)** | Python | Open-source speech recognition |
| **TTS (Coqui / Tacotron)** | Python | Text-to-speech synthesis |

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### ☁ Cloud AI SDKs (for Production Integration)

| Library | Cloud Platform | Common Use |
|---|---|---|
| **Boto3 (AWS SDK)** | AWS | Integrate AI with AWS S3, EC2, Rekognition |
| **Google Cloud AI SDK** | GCP | AutoML, Vision API, NLP API |
| **Azure AI SDK** | Azure | Cognitive Services (Vision, Speech, Language) |
| **AWS SageMaker SDK** | AWS | ML model deployment and training |
| **Vertex AI SDK** | GCP | Unified ML pipeline deployment |

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### □ Visualization & Analytics

| Library | Language | Common Use |
|---|---|---|
| **Matplotlib** | Python | 2D data visualization |
| **Seaborn** | Python | Statistical visualization |

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|---|---|---|
| **Plotly / Dash** | Python, JavaScript | Interactive analytics dashboards |
| **Bokeh** | Python | Real-time interactive plots |
| **Tableau SDK / Power BI SDK** | Python, Java | Business intelligence integration |

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 **MLOps & Model Deployment**

| Library | Language | Common Use |
|---|---|---|
| **MLflow** | Python | Model tracking, experiment management |
| **Kubeflow** | Python, YAML | ML pipelines on Kubernetes |
| **TensorFlow Serving** | Python, C++ | Model deployment service |
| **TorchServe** | Python | Deploy PyTorch models |
| **ONNX / ONNX Runtime** | Multi-language | Model conversion and optimization |
| **FastAPI / Flask** | Python | REST API for AI model serving |
| **Docker / Kubernetes** | Platform | Containerization and orchestration |

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✅ **Summary — Most Commonly Used Daily in IT Projects:**

- **Modeling:** Scikit-learn, TensorFlow, PyTorch
- **NLP:** Transformers, spaCy
- **Vision:** OpenCV, YOLO
- **Data:** Pandas, NumPy
- **Serving:** FastAPI, MLflow, Docker
- **Cloud Integration:** Boto3, Google AI SDK, Azure AI SDK

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Would you like me to provide this list in **tabular Excel/CSV format** (for documentation or team knowledge base use)?